

VIBHOR AGARWAL

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EDUCATION

University of Surrey

Doctor of Philosophy in Computer Science

July 2021 - Present

Surrey, England, UK

The LNM Institute Of Information Technology

Bachelor of Technology in Computer Science and Engineering

August 2016 - June 2020

Jaipur, India

Overall GPA: 9.45/10

RESEARCH PUBLICATIONS

GraphNLI: A Graph-based Natural Language Inference Model for Polarity Prediction in Online Debates The ACM Web Conference 2022

Vibhor Agarwal, Sagar Joglekar, Anthony P. Young, Nishanth Sastry

- Developed a novel Graph-based deep learning model to predict the argumentative relations of *attack* and *support* (polarities) in online debates.
- GraphNLI leverages neighboring context in online discussions through graph walks, along with the local context, and has wide applications in NLP field, such as Recognizing Textual Entailment.

“Way back then”: A Data-driven View of 25+ years of Web Evolution TheWebConf 2022

Vibhor Agarwal, Nishanth Sastry

- An exploratory data analysis and measurement study to quantify how the Web has evolved, by leveraging historical data crawled from *Archive.org*, *Google Trends*, and *Alexa*.

AnnoBERT: Effective Representation of Multiple Annotators’ Perspectives and Label Semantics for Hate Speech Detection Submitted to NAACL 2022

Wenjie Yin, Vibhor Agarwal, Aiqi Jiang, Arkaitz Zubiaga, Nishanth Sastry

- Model different annotators’ perspectives using Collaborative Topic Regression in subjective tasks such as Hate Speech detection.
- AnnoBERT fuses annotator perspectives and label semantics to improve Hate Speech detection.

Under the Spotlight: Web Tracking in Indian Partisan News Websites ICWSM 2021

Vibhor Agarwal, Yash Vekaria, Pushkal Agarwal, Sangeeta Mahapatra, Shounak Set, Sakthi Balan Muthiah, Nishanth Sastry, Nicolas Kourtellis

- Collected the dataset of traffic information using OpenWPM for the Indian news websites.
- Implemented various Privacy Metrics such as Canvas Fingerprinting, Cookie Synchronization, and Invisible Pixels to measure the extent of tracking across partisan news websites.

Differential Tracking Across Topical Webpages of Indian News Media WebSci 2021

Yash Vekaria, Vibhor Agarwal, Pushkal Agarwal, Sangeeta Mahapatra, Sakthi Balan, Nishanth Sastry, Nicolas Kourtellis

- Studied differential tracking across different news topics in Indian News Websites and found preferential attachment of trackers towards specific topics.
- Developed DiBETS, a novel machine learning methodology, to automatically classify webpage URLs into news topics.

A Metadata-based Event Detection Method using Temporal Herding Factor and Social Synchrony on Twitter Data

RCIS 2021

Nirmal Sivaraman, Vibhor Agarwal, Yash Vekaria, Sakthi Balan Muthiah

- Research project to analyse and detect social synchronies and thereafter, to detect online events from these detected synchronies on Twitter dataset.
- A social synchrony is a social phenomenon where many people perform a similar action in unison over a short period of time during which the number of people who perform the action first increases and then decreases.
- Dataset used: Time-series dataset of tweets on daily basis collected using Twitter API.

Video Surveillance System for Female Security using Keras

PyCon US 2020

Poster - Vibhor Agarwal

- This research project was part of India Innovation Challenge Design Contest' 2018 organised by DST, Govt. of India. Our team was selected out of 10146 teams for the contest.
- Aimed at pro-actively detecting suspicious activities like molestation against women and automates the current manual Surveillance process to enhance the security standards for women in the society.
- Used hybrid neural networks containing stacked Convolutional Neural Networks and Long Short Term Memory Networks sequentially to capture spatial and temporal information from the live video streams.

WORK EXPERIENCE

The Alan Turing Institute

Jul 2021 - Sep 2021

Visiting Researcher, Data Study Group

London, UK

- In this Data Study Group, organized by Turing Institute and University of Leeds, UK, I worked on a research project to predict Vet-AI's demands.
- Vet-AI is an AI powered research company for Pets' healthcare. Using the rich Vet-AI dataset, our group did data analysis and used time-series and machine learning techniques to predict demands for their online consultations.

Media.net (Directi)

Jan 2020 - Jun 2021

Site Reliability Engineer

Mumbai, India

- Media.Net is a leading contextual advertising tech company.
- During the six months internship (Jan-Jul' 20), I did lots of projects. The largest among them was to optimise the Machine Learning pipeline of a Keyword Recommender model and integrate it with TensorFlow Extended.
- The Keyword Recommender model, using the context of a webpage, suggests keywords for which ads can be shown on webpages.

SLK Software

May 2019 - Jul 2019

Software Engineering Internship

Bangalore, India

- Built cross-platform desktop application using Electron and ReactJS to analyse Java applications.
- Given the project path as input, the application can generate its corresponding Class Diagram and Data-Flow Diagram for analyses of the classes and properties like inheritance, etc.
- Also, finds the dead code, unused variables, duplicate imports and other problems in the project.
- Technologies used - JavaScript (ReactJS, Electron, Node.js), Python, and MongoDB.

Google Summer of Code' 2018 — The Oppia Foundation

May 2018 - Aug 2018

Open Source Developer

Remote

- Oppia aims at building interactive platform for both lesson creators and learners and brings learning freely available in the interesting and story telling way.
- Implemented two new interactions- 'Number with units' and 'Drag and drop sort' to enhance the learning experience of the students and help creators in making effective lessons.
- Technologies used - JavaScript (AngularJS, JQuery), Python, and Google Cloud.

AWARDS AND ACHIEVEMENTS

- Received **Ultimate Web Science Quiz Award** at ACM WebSci 2021.
- Received funded **PhD studentship** at University of Surrey, UK.
- Received the **Best Bachelor's Thesis Award** in the graduated class of 2020 at LNMIIT, Jaipur, India.
- Awarded **PyCon US 2020 scholarship** to attend the largest Python Conference in the world and present my poster titled, "Video Surveillance System for Female Security using Keras".
- **Quarter finalist in India Innovation Challenge Design Contest' 2018** organised by Department of Science and Technology and IIM Bangalore for our robust 'Pro-active Video Surveillance System'.
- **Awarded Meritorious Certification and Scholarship** for securing position in top 1 percentile at LNMIIT, Jaipur.
- Awarded **Certificate of Appreciation** for **Google Summer of Code' 18** at LNMIIT, Jaipur.
- Earned **Silver Medal** at **State level Mental ability competition (Abacus)**.

TECHNICAL SKILLS

Programming Languages	Python, JavaScript, C, Java, MATLAB
Artificial Intelligence	Machine Learning, Deep Learning, NLP, Computer Vision
Tools	Keras, TensorFlow, Scikit-Learn, Numpy, Pandas, Matplotlib, and Seaborn
Frameworks	Flask (Python), AngularJS, ReactJS, Electron, and Node.js (JavaScript)
Database	MySQL, SQLite and MongoDB and Redis (NoSQL)
Version Control	Git and GitHub

OPEN SOURCE

Data Version Control (DVC) Jul 2019 - Nov 2019

- DVC is an open-source version control for Data Science and Machine Learning projects.
- Mostly contributed to the technical documentation of the project and made bug fixes in the code.

The Oppia Foundation Sep 2017 - Jul 2019

- Contributed to this great Google's open source project for almost two years.
- Was part of core developer's team, implemented new features and helped new Geeks in getting started with the basics. Was also a **maintainer** keeping track of the Pull Requests.

POSITIONS OF RESPONSIBILITY

Lab Demonstrator — Natural Language Processing Lab Feb 2022 - Present

- Helping Master's students in understanding and implementing the NLP concepts using Python.

Lab Demonstrator — Computational Intelligence Lab Sep 2021 - Jan 2022

- Helped Master's students in understanding the concepts of Machine Learning and Computational Intelligence.

Data Science Lead — Computer Society of India (LNMIIT Chapter) Aug 2018 - May 2019
Computer Society of India (CSI) is the first and the largest body of computer professionals in India.

- Guided beginner 1st and 2nd year students in accomplishing the projects related to Data Science.
- Conducted various workshops and time-to-time talks related to state-of-the-art algorithms in ML and Data Science.

Google Code-In 2018 Mentor May 2018 - Aug 2018

- Selected as the mentor in GCI' 18 program to guide school students to contribute in open source projects under **JBoss (RedHat)** organization.